From the Published Construct to a Defensible Data-Generating Process: A Step-by-Step Roadmap

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Purpose. This paper sets out, in a recommended order, the changes that take the data-generating process (DGP) of Hendrickson et al. [1] (`orig`, commit `58b32a9`) to the construct this project now recommends, and beyond it to the pharmacodynamic model proposed in `docs/38`. For each step it gives:

- what the step changes and which defect it removes;
- the evidence and its status;
- what it costs, in feasible effect size, comparability and analysis;
- whether the package (`R/generateData.R`) already implements it.

The rationale for individual changes is developed elsewhere: `docs/32`, `docs/34`, `docs/35`, `docs/36`, `docs/37`, `docs/38` and `whitepaper-covar-vs-orig-dgp.md`. This paper puts them in sequence and says where the project stands.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
BR, PB, TV	drug, expectancy and natural-course responses	canonical
B_i, b_i	biomarker; standardized biomarker	canonical
c_{bm}	covariance-moderation strength (correlation of B with BR where coupled)	canonical
β_{bm}	mean-moderation multiplier	canonical
$c_{bm,TV}$, $c_{bm,PB}$	prognostic biomarker-TV and biomarker-PB correlations (`docs/46`)	local
β_{bm}^{TV}	mean moderation of the natural course by b_i	canonical (component slope)
N	total sample size across paths	canonical
ρ	within-component correlation: at every lag under CS, per week under AR(1)	canonical
c_1, $c_\times$	cross-component correlation at the same visit and at different visits (code `c.cf1t`, `c.cfct`)	local
$t_{1/2}$, λ	carryover half-life; $\lambda = \ln 2 / t_{1/2}$	canonical
t_{od}, t_{sd}	time on drug; time since discontinuation	canonical
g_t	coupling at visit t: the fraction of c_{bm} applied	local
c_{bm}^*	ceiling: the largest c_{bm} with every path's matrix positive definite	local
w_t, w_s	calendar weeks of visits t and s	local
K, A	3×3 component correlation; common time correlation of the separable block $K \otimes A$	local
μ_{stop}	BR mean at discontinuation	local
$f(t)$	population drug-response time course on $[0, 1]$ (`docs/38`)	local
F, $\bar{f}_{\text{on}}$, $\bar{f}_{\text{off}}$	contrast-weighted exposure $\bar{f}_{\text{on}} - \bar{f}_{\text{off}}$; on- and off-drug means of $f(t)$	local
ω_i, $\bar{\omega}$	participant drug sensitivity (`docs/38`'s θ_i); its population mean	local
ζ	sensitivity moderation by the biomarker (`docs/38`'s κ)	local
v_i, σ_v	unexplained sensitivity; its SD	local
ξ_{it}	serial noise in BR_{it} in the `docs/38` model	local

ω_i and ζ are renamed from `docs/38`, whose θ, κ and z collide with canonical symbols (a generic interaction coefficient, the standard-error ratio, and the standardized biomarker b_i). v_i and ξ_{it} were written u_i and ε_{it} in this document before 2026-10-07; they were renamed because NOTATION reserves u_i for the participant random intercept and ε_{it} for the observation-level residual.

- **Coupling rule.** How g_t is set. *Step* (orig): $g_t = 1$ wherever the mean of BR is nonzero. *Graded*: 1 on drug, $e^{-\lambda t_{sd}}$ off drug after exposure, 0 before any exposure.
- **CS.** Compound symmetry: correlation ρ at every lag.
- **AR(1).** Correlation $\rho^{|w_t - w_s|}$ between visits in weeks w_t, w_s.
- **Separable.** The response block is $K \otimes A$: a 3×3 component correlation K times a common time correlation A, so cross-component correlations decay with time like within-component ones (`docs/35`).
- **Configuration.** The lettered constructs of `docs/36`:
  - A, published;
  - B, AR(1) with step coupling;
  - C, separable AR(1) with graded coupling;
  - D, mean moderation;
  - E, separable AR(1) with mean moderation;
  - F, AR(1) with decaying cross-factor and graded coupling.
- **Problem 1 and problem 2.** Correlation matrices that are not positive definite and are silently repaired; the collapse of power under negligible carryover (`docs/36`).

1. Summary

1. **Eight steps take the published construct to a generative model.** They are:
   1. fail loudly instead of repairing;
   2. graded coupling;
   3. anchored carryover;
   4. AR(1) correlation;
   5. separable cross-component structure with a factored generator;
   6. choosing ρ and the strength inside the ceiling;
   7. restart from the residual response;
   8. the pharmacodynamic sensitivity model of `docs/38`.

   Steps 1 to 6 give the recommended near-term construct, configuration C. Steps 7 and 8 are a proposal awaiting implementation and validation (Section 3).
2. **The order matters.** Graded coupling alone, under compound symmetry, converts problem 2 into problem 1: the ceiling stays at 0.256 while the published effect sizes are 0.3 and 0.6. AR(1) alone leaves the step rule's

collapse in place and, on the published grid, leaves more matrices to repair (48% against 37%). Each step is placed after the steps it depends on (`docs/36`, Section 6.6).

3. **The package already implements steps 2, 3 and 4.** It does not implement step 1 fully: the Cholesky step still repairs silently. It also lacks step 5 (it uses the decaying cross-factor form, configuration F) and steps 7 and 8 (Section 4).
4. **The near-term target is configuration C.** That is separable AR(1), graded coupling and anchored carryover, generated by the factored generator, which cannot simulate an invalid model. At $\rho = 0.7$ it admits $c_{bm} = 0.45$ in every design and half-life with a margin of at least 0.02. At the published $\rho = 0.8$ the Hybrid ceiling is 0.440 (`docs/36`; ceilings recomputed for this paper).
5. **Each step needs an analysis that keeps pace.** AR(1) data need a serial-correlation or unstructured working covariance with a small-sample correction. Graded coupling and anchored carryover make an exposure-weighted drug indicator the natural analysis; the covariance-study runs now in progress evaluate both (`docs/43`). *Update (2026-10-07):* those runs are complete (`docs/45`), and a stress test on configuration C has selected the analysis (`docs/46`, `docs/49`; Section 5).

Status note (2026-10-07). The step statuses below were rechecked on 2026-10-07. The package's data-generating code is unchanged since this paper was written (`R/generateData.R` last changed on 2026-09-08, inspected with the git log), so the package column of Section 4 still holds. Configuration C, as used by the covariance study and by the stress test (`11-strawman-stress-test.R`), still keeps the published recursive carryover (inspected). The changes since 2026-10-04 are on the analysis side (Section 5) and in the stress-test driver's new prognostic mechanisms (Section 4).

2. The starting point: orig

The published construct (`docs/36`, Section 3; `docs/40`, Sections 3 to 6):

Table 2: Features of the published orig construct

Feature	orig
Joint distribution	one 26-variable multivariate normal per path
Within-component correlation	CS, $\rho = 0.8$ at every lag
Cross-component correlation	$c_1 = 0.2$ same visit, $c_\times = 0.1$ otherwise
Biomarker coupling	step rule: c_{bm} wherever the BR mean is nonzero
Carryover on the BR mean	recursive, cumulative t_{sd}, over-decays
Re-exposure	time on drug restarts at zero

Feature	orig
Off-drug BR variance	full (SD 8) at every visit
Invalid matrices	silently repaired (`make.positive.definite`)

Its two known defects share one cause, the step rule. Without carryover the coupling has an on/off contrast, and the matrix is invalid for any c_{bm} above 0.256. With any carryover the coupling is constant, the matrix is valid, and the interaction disappears (`docs/36`, Section 5.4; `docs/40`, Section 5).

3. The steps

Step 1: fail loudly instead of repairing

Change. Remove every silent repair. A cell whose matrix is not positive definite stops, or is reported as infeasible, instead of being simulated from a nearby matrix.

Defect removed. The repair changes the model that is simulated. At $c_{bm} = 0.6$ the on-drug correlation actually simulated is 0.54 to 0.57, off-drug visits acquire 0.03 to 0.04, and other correlations move by up to 0.10 (`docs/36`, Section 4.3). The published no-carryover results at 0.6 describe a model other than the one stated.

Cost. Some published cells become infeasible. That is a statement about the construct, not a loss.

Package status: partial. `validateParameterGrid()` checks feasibility. Repairs are still silent in two places:

- with `makePositiveDefinite = TRUE` the repair warns only when `verbose = TRUE`;
- whatever that setting, the Cholesky factorization falls back to `make.positive.definite()` on failure, so a matrix that is not positive definite is always simulated.

The project's quick-sim drivers (`04-power-simulation.R`) skip infeasible cells instead.

Step 2: graded coupling

Change. Couple the biomarker to BR in proportion to the drug effect that remains: $g_t = 1$ on drug, $e^{-\lambda t_{sd}}$ off drug after exposure, and 0 before any exposure.

Defect removed. Problem 2. Power now declines smoothly and monotonically with the half-life, and at the published half-lives (0.1 and 0.2 weeks) the loss is under 0.01 (`docs/37`, Proposition 3, closed form validated by simulation).

Cost. Under compound symmetry the ceiling barely moves (0.256 without carryover, 0.307 at one week in Hybrid), so above 0.256 graded coupling alone turns problem 2 into problem 1 in every cell. This step must be followed by steps 4 and 5 if larger effects are to be studied.

Package status: implemented (`buildSigma`, coupling $c_{bm}e^{-\lambda t_{sd}}$ off drug; pre-exposure visits keep 0).

Step 3: anchored carryover on the drug-response mean

Change. Off drug, decay the BR mean from its value at discontinuation, $\mu_{\text{stop}}\, 2^{-t_{sd}/t_{1/2}}$, instead of recursively decaying the previous visit's already-decayed mean with a cumulative t_{sd}.

Defect removed. The recursion counts elapsed time twice, so the mean decays faster than the stated half-life. At $t_{1/2} = 1$ week, two weeks after stopping, the published mean is 1.27 where anchored decay gives 2.55 (`docs/40`, Section 3; `docs/32`, Section 6).

Cost. Results at a nominal half-life differ from the published ones, since the stated half-life now means what it says.

Package status: implemented (`buildSigma`, the `last_on` anchor). The quick-sim reproductions of orig keep the recursion deliberately, and so do the configuration C drivers (`04-power-simulation.R`, `08-covariance-study.R`, `11-strawman-stress-test.R`; inspected 2026-10-07).

Step 4: AR(1) within-component correlation

Change. Within each component, correlation $\rho^{|w_t - w_s|}$, with ρ per week, in place of a constant ρ at every lag.

Defect removed. Compound symmetry says that symptoms one week apart and sixteen weeks apart are equally correlated, which is implausible for symptom trajectories. It is also the source of the low ceiling: under CS the smallest eigenvalue of the response block is $(1 - \rho) - (c_1 - c_\times) = 0.1$ for every design (`docs/36`, Section 4.2).

Cost.

- **Applied with the published cross-component form, AR(1) helps little and can hurt.** It raises the ceiling where the coupling has an on/off contrast and lowers it where the coupling is constant (OL falls from 0.893 to 0.576). On the published grid it leaves more matrices to repair (48% against 37%).

- **The analysis must change too.** On AR(1) data the published random-intercept analysis rejects 14% to 23% of null crossover trials (`docs/36`, Section 6.8; covariance study, 0.14 to 0.16).

Package status: implemented (`buildSigma`: `rho^tg` within components, and the decaying cross-component form $c_{\times}\rho^{|w_t - w_s|}$, configuration F).

Step 5: a separable cross-component structure and the factored generator

Change. Make the response block separable, $K \otimes A$, with cross-component correlations $c_1\rho^{|w_t - w_s|}$ that decay with time exactly as the within-component ones do. Generate data from the factorization, without forming the full matrix:

1. three AR(1) recursions;
2. a 3×3 mixing step;
3. a conditional draw of the biomarker (`docs/36`, Section 6.7).

Defect removed.

- **The response block is valid for every schedule.** A Kronecker product of valid matrices is valid. The decaying cross-factor form of step 4 has no such guarantee: its smallest eigenvalue falls to 0.010.
- **The ceiling rises.** The Hybrid ceiling without carryover goes from 0.291 (configuration F) to 0.440 at $\rho = 0.8$ (`docs/35`; `docs/36`, Section 6.4).
- **An infeasible effect size stops the simulation instead of being repaired.** The factored generator's conditional variance is positive exactly below the ceiling.

Cost. Separability changes feasibility, not power: with graded coupling, configurations C and F give the same power within ± 0.04 once each test is judged against its own null (`docs/36`, Section 6.8). It is a modeling assumption: all components share one time correlation.

Package status: not implemented. The package uses configuration F. Configuration C exists in the quick-sim drivers (`04-power-simulation.R`, arm C) and was used by the covariance study.

Step 6: choose the correlation and the effect size inside the ceiling

Change. A decision, not code: set ρ and the grid of c_{bm} so that every cell is feasible with a margin.

Evidence. Exact graded-coupling ceilings, minimum over designs, recomputed for this paper (`ceilings-graded-by-rho.csv`):

Table 3: Exact graded-coupling ceilings by ρ for CS, AR(1) and separable AR(1) constructs

ρ	CS	AR(1), decaying cross (F)	Separable AR(1) (C)
0.8	0.256	0.291	0.440
0.7	0.346	0.452	0.474
0.6	0.410	0.457	0.462
0.5	0.458	0.448	0.448

These are the binding cells, at $t_{1/2} = 0$. At $\rho = 0.8$ and 0.7 carryover raises or keeps every ceiling. At lower ρ a few long-half-life cells fall slightly: at $\rho = 0.5$ the CO ceilings at $t_{1/2} = 2$ are 0.003 to 0.004 below their no-carryover values, under both AR(1) forms (verified in `ceilings-graded-by-rho.csv`, 2026-10-07; the earlier sentence said carryover raises every ceiling). The minimum over designs is still at $t_{1/2} = 0$ in every row of the table.

Recommendation. Configuration C at $\rho = 0.7$ admits $c_{bm} = 0.45$, the compendium's reference value (NOTATION.md, rule 2), everywhere with a margin of at least 0.02. Alternatively keep the published $\rho = 0.8$ and cap c_{bm} at 0.4. Avoid running within about 0.02 of a ceiling: there the conditional variance of BR given the biomarker approaches zero and results become fragile.

Cost. ρ is itself an unestimated assumption about PTSD symptom autocorrelation. Changing it changes power at a given strength, so results across ρ are not directly comparable.

Step 7: restart from the residual response on re-exposure

Change. When the drug is restarted, let the response rebuild from its residual level, not from zero. The simplest version keeps the Gompertz onset curve and starts it at the level corresponding to the residual response. The principled version is the turnover model of step 8 (`docs/38`, Section 3).

Defect removed. Time on drug resets at every re-exposure, so the Hybrid crossover's four-week on-drug period recovers only 39% of the ceiling (4.3 of 11 points) even when the previous exposure ended weeks earlier with a near-full response (`docs/40`, Section 4). For a drug whose benefit persists, that understates the crossover's on-drug response.

Cost. It interacts with carryover: a long response half-life raises both the residual off-drug level and the restart level. The pharmacology is uncertain, so the step should be reported as a sensitivity, not imposed.

Package status: not implemented (`buildtrialdesign` resets t_{od}).

Step 8: the pharmacodynamic sensitivity model (`docs/38`)

Change. Replace the matrix specification of BR with a generative model:

$$BR_{it} = \omega_i \, f(t) + \xi_{it}, \qquad \omega_i = \bar{\omega}\,(1 + \zeta b_i) + v_i,$$

with $f(t)$ the turnover time course (anchored decay, restart from the residual), ω_i the participant's drug sensitivity moderated by the standardized biomarker, v_i unexplained sensitivity, and ξ_{it} serial noise. The mean, the variances and every correlation follow (`docs/38`, Section 5).

Defects removed.

- **Moderation is proportional to the response:** small early in titration, decaying with the response off drug, never a step.
- **The off-drug drug-response variance shrinks with the response.** Every current construct gives BR its full SD of 8 at off-drug visits with zero mean. `docs/38` judges this the least defensible feature of the construct, and it adds noise to every on-off contrast.
- **Validity by construction.** The covariance is a rank-one sensitivity term plus a valid noise covariance. There is no ceiling on effect size, because correlations are outputs, not inputs.

Consequences for the interaction test (`docs/38`, Section 6; derived, not simulated). Power is strictly increasing in the contrast-weighted exposure $F = \bar{f}_{\text{on}} - \bar{f}_{\text{off}}$. It is capped by the share of sensitivity variation the biomarker explains. And the contrast variance now depends on carryover.

Cost.

- **A new architecture.** All results through `generateData()` would need comparison before any are replaced.
- **New parameters:** $\bar{\omega}$, ζ (or the peak correlation), σ_v, the serial noise, and the response half-life. The last is unknown for prazosin in PTSD.
- **A modeling choice:** multiplicative moderation assumes the biomarker acts on sensitivity. A biomarker that acted on onset or offset rates would need a different model.

Package status: not implemented; proposal. Required checks before use (`docs/38`, Section 9):

- size holds when $\zeta = 0$;
- results reduce to the no-carryover case as $t_{1/2} \to 0$;
- simulated moments match the derivations.

4. Status and the recommended target

Table 4: Status of each roadmap step for orig, covar, Configuration C and the docs/38 model

Step	orig	Package (`covar`)	Configuration C (quick-sim)	`docs/38` model
1 Fail loudly	no	partial (Cholesky fallback repairs)	yes (cells skipped)	yes (no matrix)
2 Graded coupling	no (step)	yes	yes	yes (proportional)
3 Anchored carryover	no	yes	no (recursion kept)	yes
4 AR(1)	no (CS)	yes	yes	yes (serial noise)
5 Separable, factored	no	no (F)	yes	not needed
6 ρ, strength in range	0.3 and 0.6 exceed 0.256	grid-dependent	$\rho = 0.7$, 0.45	no ceiling
7 Restart from residual	no	no	no	yes
8 Sensitivity model	no	no	no	yes
Hybrid ceiling, $t_{1/2} = 0$	0.256	0.291 at $\rho = 0.8$	0.440 at 0.8; 0.480 at 0.7	none

Near-term target: configuration C with anchored carryover, at a chosen ρ. This needs four changes:

1. **Add the separable form to `buildSigma`**, as a new value of the cross-component option alongside the current decaying form.
2. **Add the factored generator** as the sampler for the separable form. It stops on an infeasible effect size.
3. **Remove the Cholesky fallback,** or make it an error.
4. **Use anchored carryover in configuration C.** The quick-sim version keeps the published recursion for comparability with orig; the package version already anchors.

Status (2026-10-07): none of the four changes has been made; the table above was rechecked against the package and the configuration C drivers on this date and is unchanged (inspected).

Longer-term target: the `docs/38` model, implemented as a new `dgp_architecture` value after the checks of step 8. It would be run alongside configuration C until the two have been compared.

Branches that remain available. Mean moderation (`'mean_moderation'`, constant or trajectory-scaled), the dual-channel architecture (`'combined'`, manuscript 11), the contaminated-biomarker option (`c.bm.pb`, manuscript 06), reduced component sets (manuscripts 13 and 14) and alternative response-curve families (manuscript 07) are orthogonal to steps 1 to 6 and can be combined with configuration C. One extension is missing: there is no parameter for a biomarker related to natural course, so a biomarker that predicts TV cannot be simulated (`docs/31`, Section 18.1). *Update (2026-10-07):* this is still true of the package, but the stress-test driver (`11-strawman-stress-test.R`) now simulates prognostic biomarkers on configuration C: a constant biomarker-TV correlation $c_{bm,TV}$, a mean moderation of TV that grows along its curve (β_{bm}^{TV}), a biomarker-PB correlation $c_{bm,PB}$, and a biomarker-baseline correlation (`docs/46`, Section 5; inspected). These are candidates for the package.

5. Analysis that keeps pace with the DGP

Table 5: Analysis implications and supporting evidence for each DGP step

DGP step	Analysis implication	Evidence
2, 3 (graded coupling, anchored carryover)	an exposure-weighted drug indicator `Dbc` matches the coupling's decay; binary `Db` remains valid but loses power under carryover	`docs/37`; manuscript 02 (Hybrid, OL+BDC); `docs/45`, Section 7 (the `Dbc` run, completed)
4 (AR(1))	the working covariance must allow serial correlation: `corCAR1` or a structured covariance with CR2; the unstructured covariance with Kenward-Roger is slightly liberal and failed the stress test	covariance study (`docs/45`); stress test (`docs/46`); manuscripts 02 and 10
5, 6 (feasible strengths)	larger effects become testable; power comparisons should be made at feasible, matched strengths	`docs/36`, Section 6.8

DGP step	Analysis implication	Evidence
7, 8 (pharmacodynamic model)	`Dbc` built from the response half-life; a nonlinear turnover model could estimate that half-life	`docs/38`, Section 8

Update (2026-10-07). The table's evidence column was revised on this date. The exploratory stress test on configuration C (`docs/46`; 41 cells, 28,000 trials, 14 analyses) selects F1: a random intercept with `corCAR1` residuals and CR2 inference, the baseline row, linear time, `Dbc` at an assumed half-life of 0.5 week, and the expectancy term `De` with its biomarker interaction (`docs/49`). The unstructured covariance with Kenward-Roger, which `docs/45` recommended, rejected 0.057 of null trials pooled and 0.087 at $N = 35$, and failed all three original prognostic cells. The prognostic cells are the reason the analysis needs the expectancy term: they are DGP extensions the package does not yet have (Section 4).

A coding warning for `Dbc`. Off-drug visits before any exposure must be coded 0, not $e^{-\lambda \cdot 0} = 1$. Manuscript 02's implementation codes them 1, which gives CO's placebo-first path `Dbc` = 1 at every visit. That plausibly explains its finding that exposure weighting is markedly inferior in CO (inspected in `analysis/scripts/carryover-sensitivity/simulation-core.R`; to be corrected and rerun). *Update (2026-10-07):* `docs/45` (Section 8) confirmed the effect on the same data (CO power 0.484 to 0.744 under CS). The driver has since been corrected: `exposure_weight()` now codes pre-exposure off-drug visits 0 (inspected). A rerun of manuscript 02's CO cells was not confirmed in this review.

6. Comparability with the published results

Every step departs from the published construct, so no corrected result can be set directly beside a published figure. Keep a faithful reproduction of orig as a reference arm (the published reference arm, configuration A without repair; `docs/36` called it the strawman, a word `docs/46` now uses for a provisional analysis), report every corrected result alongside it, and attribute each difference to a named step. Steps 1, 2 and 3 correct errors and need no defense beyond the evidence cited. Steps 4 to 8 are modeling choices and should be presented as such.

7. Evidence status and limitations

- **Verified.** The ceilings (exact computations, recomputed for this paper; Step 6 table rechecked against `ceilings-graded-by-rho.csv` on 2026-10-07); the effects of the repair, the step rule, AR(1) and separability

(`docs/36`); the closed-form results of `docs/37`; the covariance-study results.

- **Inspected.** Package status, read in `R/generateData.R` at the current working tree (rechecked 2026-10-07: the Cholesky fallback to `make.positive.definite()` is still present, and there is no separable cross-component option); the manuscript 02 `Dbc` coding.
- **Derived, not simulated.** Step 8 and its consequences.
- **Not evaluated.** Step 7 (time on drug restarts at zero on re-exposure in every current DGP); the combination of anchored carryover with configuration C.
- **Limitations.**
  - The roadmap concerns the covariance-moderation architecture. Mean moderation needs only steps 3, 4, 7 and 8.
  - The choice of ρ and of the response half-life are open empirical questions for prazosin in PTSD.

8. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013.

Project documents: `docs/31`, `docs/32`, `docs/34`, `docs/35`, `docs/36`, `docs/37`, `docs/38`, `docs/40`, `docs/43`, `whitepaper-covar-vs-orig-dgp.md`.